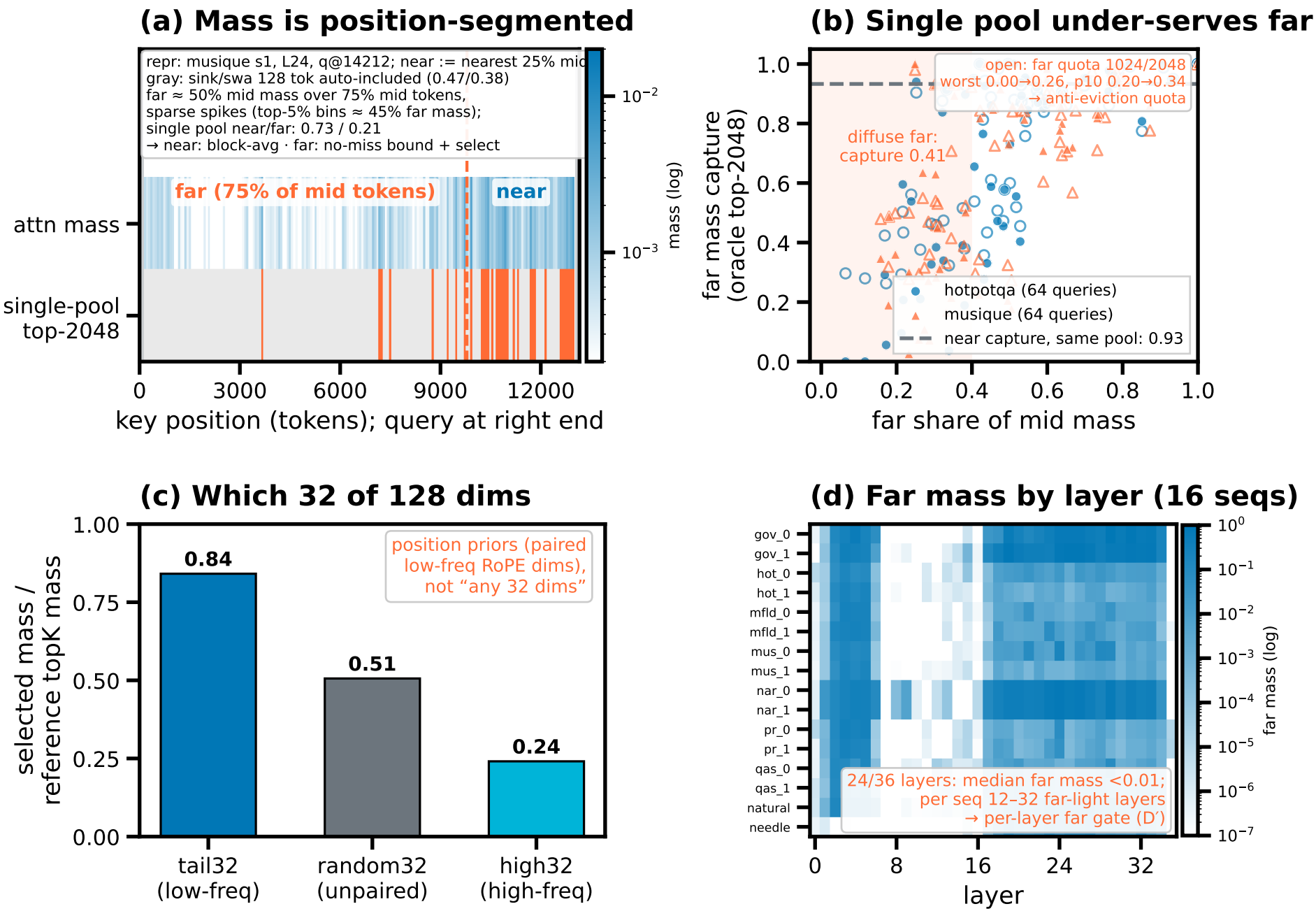

(a) Mass is position-segmented
repr: musique s1, L24, q@14212; near := nearest 25% mid
gray: sink/swa 128 tok auto-included (0.47/0.38)
far ≈ 50% mid mass over 75% mid tokens,
sparse spikes (top-5% bins ≈ 45% far mass);
single pool near/far: 0.73 / 0.21
→ near: block-avg · far: no-miss bound + select
far (75% of mid tokens)
near
attn mass
single-pool top-2048
0
3000
6000
9000
12000
key position (tokens); query at right end
10^{-2}
10^{-3}
mass (log)
(b) Single pool under-serves far
open: far quota 1024/2048
worst 0.00→0.26, p10 0.20→0.34
→ anti-eviction quota
diffuse far:
capture 0.41
hotpotqa (64 queries)
musique (64 queries)
near capture, same pool: 0.93
far mass capture
(oracle top-2048)
far share of mid mass
0.0
0.2
0.4
0.6
0.8
1.0
(c) Which 32 of 128 dims
position priors (paired
low-freq RoPE dims),
not “any 32 dims”
0.84
0.51
0.24
selected mass /
reference topK mass
tail32
(low-freq)
random32
(unpaired)
high32
(high-freq)
(d) Far mass by layer (16 seqs)
gov_0
gov_1
hot_0
hot_1
mfld_0
mfld_1
mus_0
mus_1
nar_0
nar_1
pr_0
pr_1
qas_0
qas_1
natural
needle
24/36 layers: median far mass <0.01;
per seq 12–32 far-light layers
→ per-layer far gate (D′)
0
8
16
24
32
layer
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
far mass (log)